베이지안 기계학습
6. Bayesian networks

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Bayesian networks

Bayesian networks

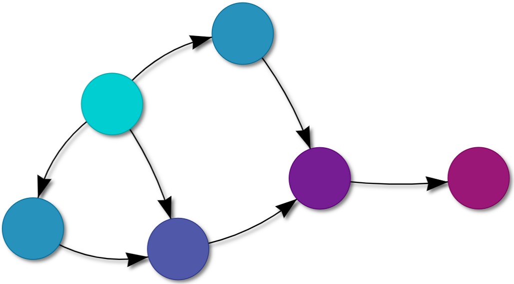


Figure: from https://gradientdescending.com/simulating-data-with-bayesian-networks/

- A Bayesian network (BN) is a probabilistic graphical model that represents a joint probability distribution over a set of random variables.

Bayesian networks

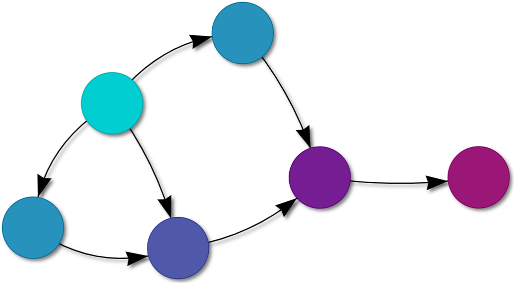
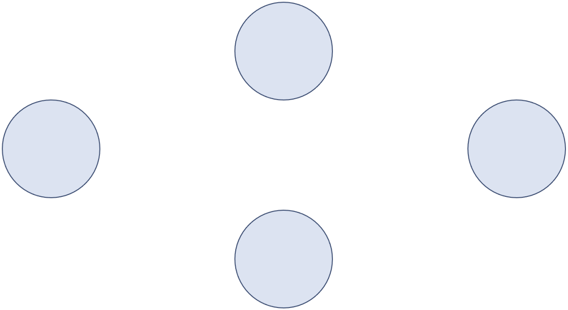


Figure: from `https://gradientdescending.com/simulating-data-with-bayesian-networks/`

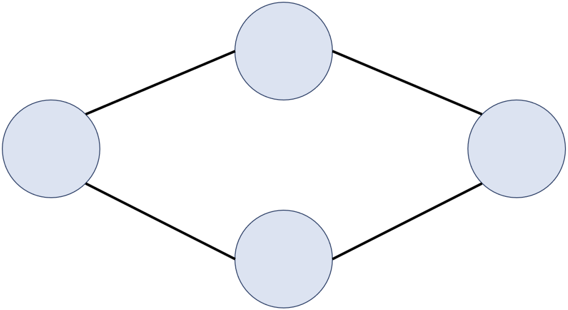
- A Bayesian network uses a directed acyclic graph (DAG) and a set of conditional probability distributions.

*Directed acyclic graph (DAG)



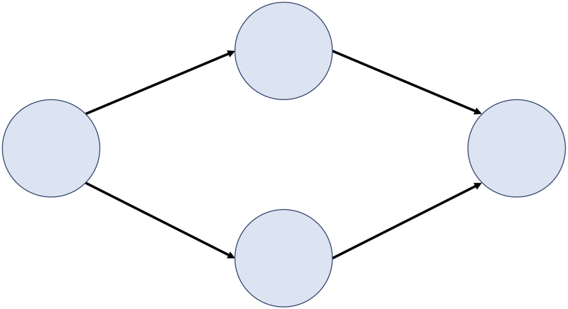
- (Vertices) $V = \{1, \ldots, p\}$

*Directed acyclic graph (DAG)



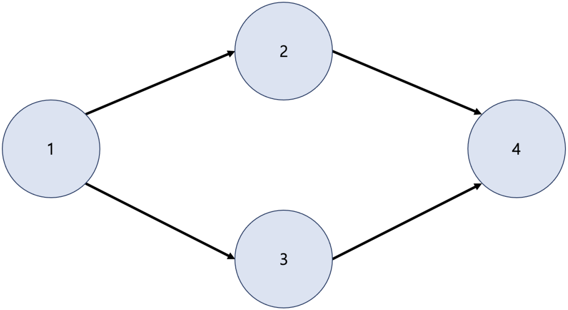
- (Edges) $E \subset V \times V$.

*Directed acyclic graph (DAG)



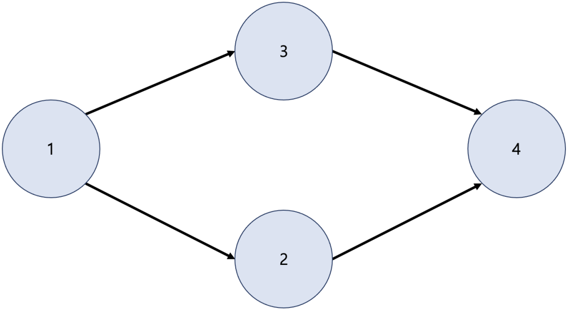
- An edge is called directed if $(i, j) \in E$ implies $(j, i) \notin E$. Here, $(i, j) \in E$ implies $i \to j$, and we call i a parent of j.

*Directed acyclic graph (DAG)



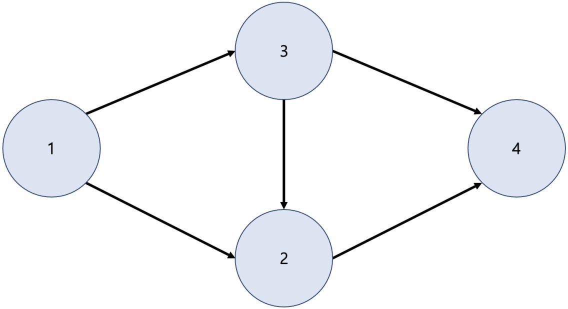
- For a DAG, there exists a parent ordering $\pi = (\pi_1, \ldots, \pi_p)$.
- A parent ordering represents the directions of edges such that for every directed edge $(j, k) \in E$, j comes before k in the ordering.

*Directed acyclic graph (DAG)



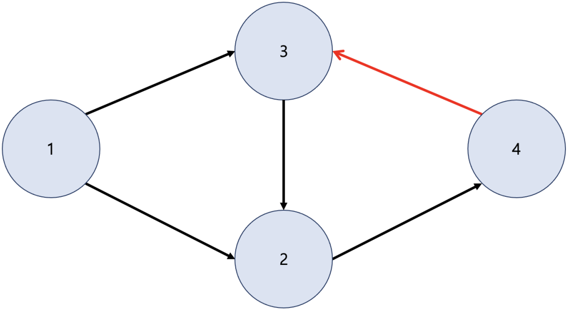
- A parent ordering is not unique.

*Directed acyclic graph (DAG)



- This is still a DAG.

*Directed acyclic graph (DAG)



- This is not a DAG! (We have a “cycle”)

*DAG models

- Let $G = (V, E)$ a DAG for a random vector $X = (X_1, \ldots, X_p)$, that is, $V = \{1, \ldots, p\}$.
- For any node $j \in V$, $\Pr(X_j \mid X_S)$ denotes the conditional distribution of the variable X_j given a random vector X_S.
- Let $pa_j(G)$ be the set consisting of the j's parents.

*DAG models

- Then, a DAG model has the following joint probability density function:

$$\Pr(X_1, X_2, ..., X_p) = \prod_{j=1}^{p} \Pr(X_j \mid X_{pa_j(G)}),$$

where $\Pr(X_j \mid X_{pa_j(G)})$ is the conditional distribution of X_j given its parent variables $X_{pa_j(G)} = (X_k)_{k \in pa_j(G)}$.

Linear Bayesian networks

- A linear Bayesian network is defined as follows:

$$(X_1, \ldots, X_p)^T = B(X_1, \ldots, X_p)^T + (\epsilon_1, \ldots, \epsilon_p)^T,$$

  which implies

$$X_j = \sum_{k=1}^{p} b_{jk} X_k + \epsilon_k, \quad j = 1, \ldots, p.$$

- $B = (b_{jk}) \in \mathbb{R}^{p \times p}$ is an edge weight matrix, where $b_{jj} = 0$. Note that in a parent ordering, B is a lower triangular matrix.
- ϵ_j's are independent errors with $\mathrm{E}(\epsilon_j) = 0$ and $\mathrm{Var}(\epsilon_j) = d_j$.

Linear Bayesian networks

- Let $X = (X_1, \ldots, X_p)^T$, then we have $X = BX + \epsilon$.
- Let $D = \text{diag}(d_1, \ldots, d_p)$.
- If $\Sigma = \mathrm{Var}(X)$ is the covariance matrix of X, then

$$\Sigma = (I_p - B)^{-1} D (I_p - B)^{-T}.$$

- If $\Omega = \Sigma^{-1}$ is the precision matrix of X, then

$$\Omega = (I_p - B)^T D^{-1} (I_p - B).$$

Remark

- Recovering a DAG can be decomposed into learning (i) the ordering and (ii) the presence of edges.
- (i) corresponds to the estimation of a parent ordering.
- (ii) corresponds to the estimation of the edge weight matrix B.
- In this lecture, we will focus only on (ii).

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Gaussian DAG models

*Gaussian DAG models

- Let $G = (V, E)$ denotes a DAG in a parent ordering.
- Due to the parent ordering assumption, $i \in pa_j(G)$ implies $i < j$.
- If we consider a linear Bayesian network, recall that $(I_p - B)X = \epsilon = (\epsilon_1, \ldots, \epsilon_p)^T \sim N_p(0, D)$.
- Thus, $X_1 = \epsilon_1$ and

$$X_j \quad = \quad \sum_{l \leq j-1} b_{jl} X_l + \epsilon_j, \quad j = 2, \ldots, p.$$

Gaussian DAG models

- It is said that $N_p(0, \Omega^{-1})$ is a Gaussian DAG model over G if

$$b_{jl} \neq 0 \text{ if and only if } l \in pa_j(G),$$

which implies

$$\begin{aligned} X_1 &= \epsilon_1, \\ X_j &= \sum_{l \in pa_j(G)} b_{jl} X_l + \epsilon_j, \quad j = 2, \ldots, p. \end{aligned}$$

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Connection to precision matrix estimation

Estimation of the covariance/precision matrix

- The estimation of the covariance (or its inverse) matrix is crucial to reveal the dependence structure.
- Many statistical methods require the estimated covariance matrix as the starting point of the analysis (e.g. LDA and QDA).
- Estimation of the covariance/precision matrix is one of important and challenging tasks, especially when the number of variables is much larger than the sample size ($p \gg n$).

*Estimation of a large matrix

- For the consistent estimation of the high-dimensional covariance or precision matrices, various restrictive matrix classes have been proposed to reduce the number of effective parameters:
  - bandable matrices,
  - sparse matrices and
  - low-dimensional structural matrices.

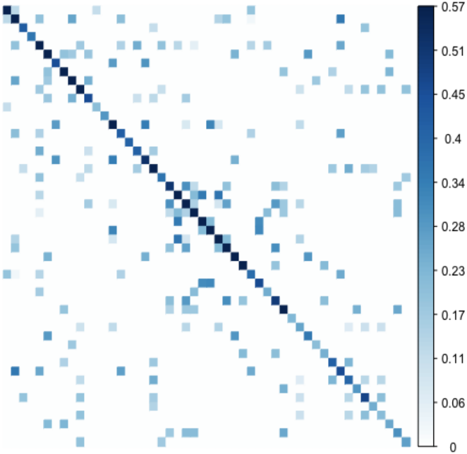

Figure: A sparse matrix.

Modified Cholesky decomposition

- Let Ω be a $p \times p$ precision matrix.
- The modified Cholesky decomposition (MCD) guarantees that there exist unique lower triangular matrix $B = (b_{jl})$ and diagonal matrix $D = diag(d_j)$ such that

$$\Omega = (I_p - B)^T D^{-1} (I_p - B),$$

  where $b_{jj} = 0$ and $d_j > 0$ for all $j = 1, \ldots, p$.
- We call B the Cholesky factor.

Advantages

- A MCD-based approach has two advantages.
  - Positive definiteness of $\Omega = (I_p - B)^T D^{-1} (I_p - B)$.
  - A sequence of regression models interpretation:

$$X = (X_1, \ldots, X_p)^T \mid \Omega \quad \sim \quad N_p(0, \Omega^{-1})$$

$$\iff \quad \begin{cases} X_1 \mid d_1 \sim N(0, d_1), \\ X_j \mid X_{1:(j-1)}, b_j, d_j \sim N\Big(\sum_{l=1}^{j-1} b_{jl} X_l,\ d_j \Big),\ j = 2, \ldots, p \end{cases}$$

where $b_j = (b_{j1}, \ldots, b_{j,j-1})^T$ is the first $(j-1)$ elements of the jth row of B.

*Sparse matrices

- The sparsity pattern can be encoded in many different ways which lead to different graph models:
  - sparse covariance matrices (Σ),
  - sparse precision matrices ($\Omega = \Sigma^{-1}$) or
  - sparse Cholesky factors (B, where $\Omega = (I_p - B)^T D^{-1} (I_p - B)$).
- In this lecture, we focus on imposing sparsity on the Cholesky factor of a precision matrix, which corresponds to a directed acyclic graph (DAG) model.

Setting & goal

- $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N_p\left(0, \Omega^{-1}\right)$
- Assume p can be much larger than n, possibly $p \gg n$.
- We assume a sparse Cholesky factor.
- The main goals are
  - support recovery of the Cholesky factor B,
  - estimation of the Cholesky factor B and
  - estimation of the precision matrix Ω.

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Empirical Sparse Cholesky (ESC) prior

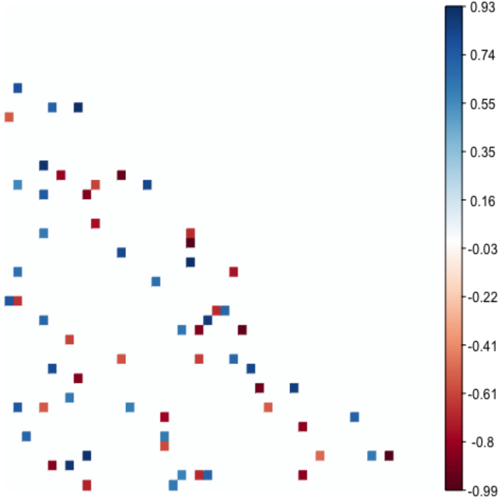
Gaussian DAG models in a parent ordering

$$X_1, \ldots, X_n \mid \Omega \overset{i.i.d.}{\sim} N_p(0, \Omega^{-1}) \quad \text{with} \quad \Omega = (I_p - B)^T D^{-1} (I_p - B)$$

$$\iff \quad \begin{cases} \tilde{X}_1 \mid d_1 \sim N_n(0,\ d_1 I_n), \\ \tilde{X}_j \mid \mathbf{X}_{S_j}, b_{S_j}, d_j, S_j \overset{ind.}{\sim} N_n\Big(\mathbf{X}_{S_j} b_{S_j},\ d_j I_n\Big), \quad j = 2, \ldots, p \end{cases}$$

- $\tilde{X}_j \in \mathbb{R}^n$ is the jth column of the data matrix $\mathbf{X}_n \in \mathbb{R}^{n \times p}$.
- $\mathbf{X}_{S_j} \in \mathbb{R}^{n \times |S_j|}$ consists of the S_jth columns of $\mathbf{X}_n$.
- $b_{S_j} \in \mathbb{R}^{|S_j|}$ is the S_jth elements of $b_j = (b_{j1}, \ldots, b_{j,j-1})^T$.
- We will impose a prior on B and D.

Prior for the Cholesky factor B (Sketch)



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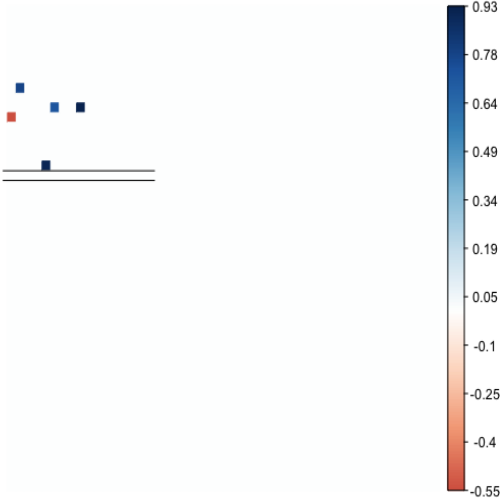

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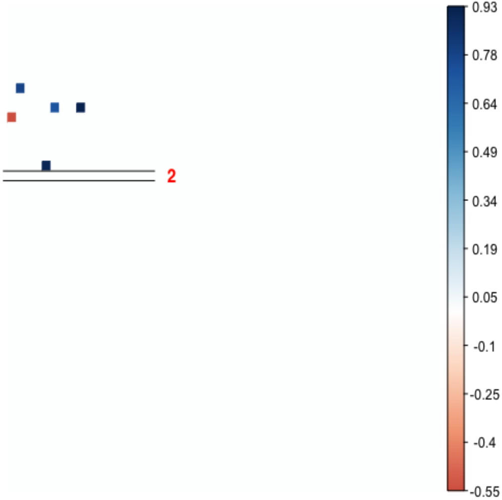

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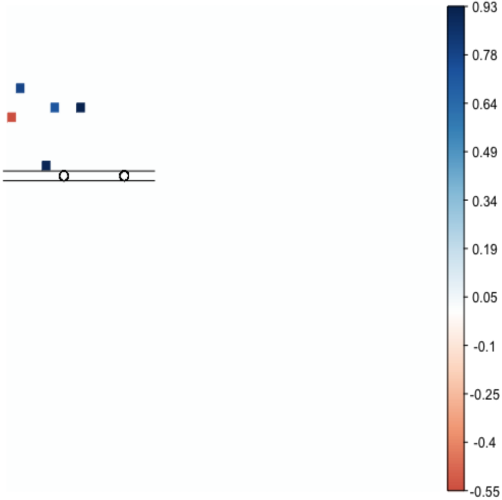

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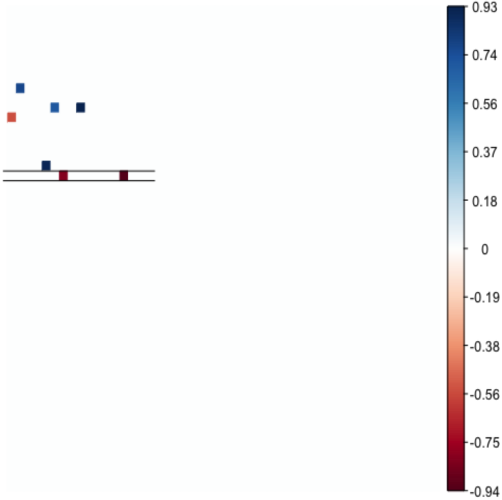

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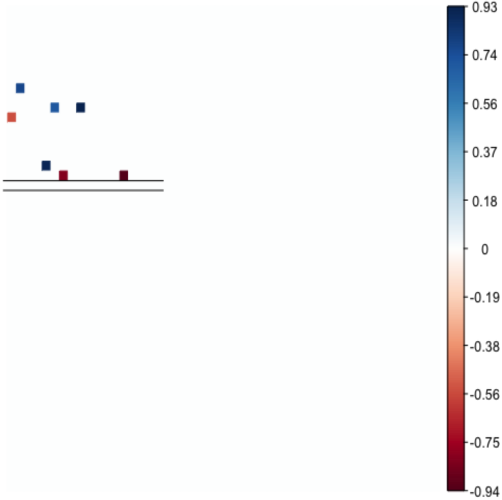

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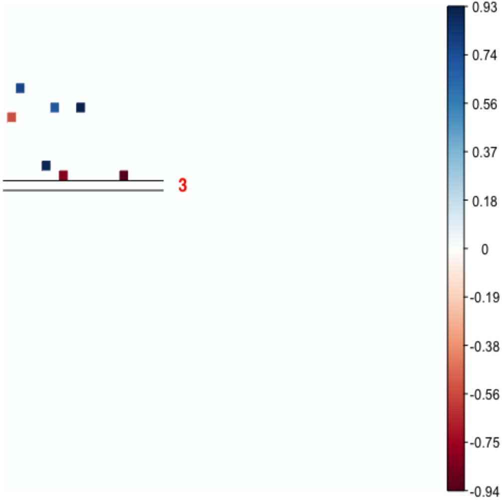

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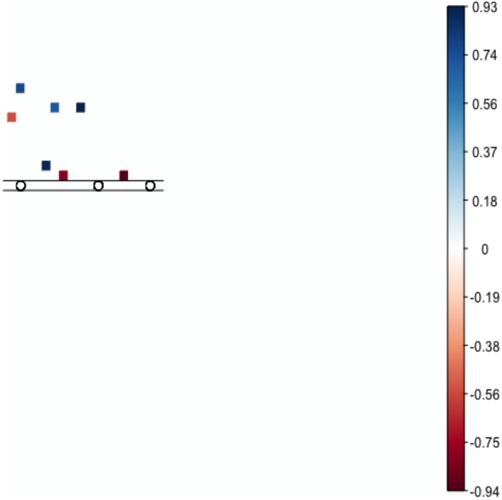
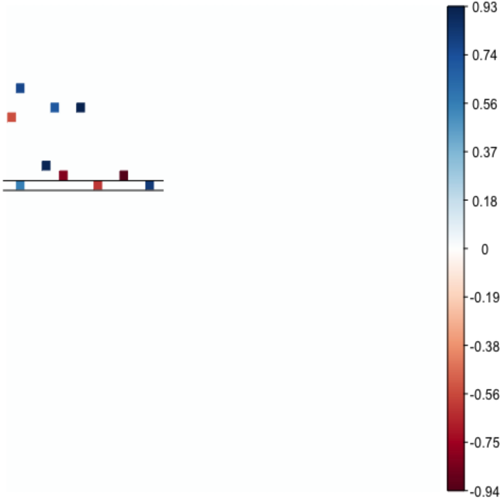


Prior for the Cholesky factor B (Sketch)



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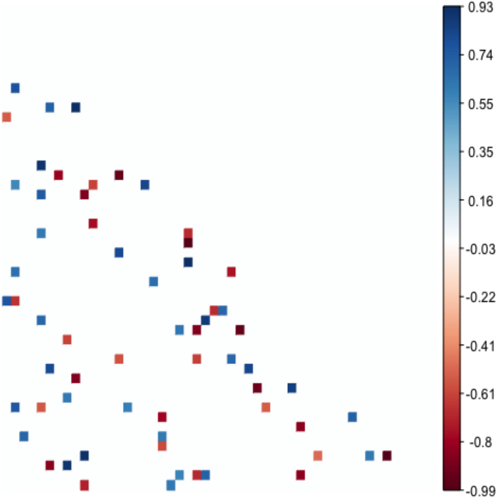

Empirical Sparse Cholesky (ESC) prior

$$\pi(B, D) \quad = \quad \pi(d_1) \prod_{j=2}^{p} \pi(b_{S_j} \mid S_j, d_j)\, \pi_j(S_j)\, \pi(d_j)$$

(i) Select the nonzero elements in B:

$$\pi_j(S_j) \quad \propto \quad f_{nj}(|S_j|) \times \binom{j-1}{|S_j|}^{-1}, \;\; j = 2, \ldots, p, \;\; S_j \subseteq \{1, \ldots, j-1\},$$

where $|S_j|$ is the cardinality of S_j and

$$f_{nj}(|S_j|) \quad \propto \quad c_1^{-|S_j|} p^{-c_2|S_j|}\, I(0 \le |S_j| \le R_j \wedge (j-1)), \qquad j = 2, \ldots, p.$$

Empirical Sparse Cholesky (ESC) prior

$$\pi(B, D) \quad = \quad \pi(d_1) \prod_{j=2}^{p} \pi(b_{S_j} \mid S_j, d_j)\, \pi_j(S_j)\, \pi(d_j)$$

(ii) For the nonzero elements in b_j, impose a version of the Zellner's g-prior:

$$b_{S_j} \mid d_j, S_j \quad \overset{ind.}{\sim} \quad N_{|S_j|}\left(\widehat{b}_{S_j},\ \frac{d_j}{\gamma}\left(\mathbf{X}_{S_j}^T \mathbf{X}_{S_j}\right)^{-1}\right),\ \ j = 2, \ldots, p,$$

where $\widehat{b}_{S_j} = \left(\mathbf{X}_{S_j}^T \mathbf{X}_{S_j}\right)^{-1} \mathbf{X}_{S_j}^T \tilde{X}_j$.

(Recall) $\quad \tilde{X}_j \mid \mathbf{X}_{S_j}, b_{S_j}, d_j, S_j \quad \overset{ind.}{\sim} \quad N_n\left(\mathbf{X}_{S_j} b_{S_j},\ d_j I_n\right), \quad j = 2, \ldots, p.$

Empirical Sparse Cholesky (ESC) prior

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where $\widehat{b}_{S_j} = \big(\mathbf{X}_{S_j}^T \mathbf{X}_{S_j}\big)^{-1} \mathbf{X}_{S_j}^T \tilde{X}_j$.

(Recall) $\qquad \tilde{X}_j \mid \mathbf{X}_{S_j}, b_{S_j}, d_j, S_j \overset{ind.}{\sim} N_n\Big(\mathbf{X}_{S_j} b_{S_j},\ d_j I_n\Big), \quad j = 2, \ldots, p.$

Empirical Sparse Cholesky (ESC) prior

$$\pi(B, D) = \pi(d_1) \prod_{j=2}^{p} \pi(b_{S_j} \mid S_j, d_j)\, \pi_j(S_j)\, \pi(d_j)$$

(iii) Impose priors for d_j's:

$$d_j \overset{i.i.d.}{\sim} IG\Big(\frac{\nu_0}{2}, \frac{\nu_0'}{2}\Big), \quad j = 1, \ldots, p.$$

*Posterior computation

The induced posterior has a closed form:

$$
\begin{aligned}
b_{S_j} \mid d_j, S_j, \mathbf{X}_n &\overset{ind}{\sim} N_{|S_j|}\left(\widehat{b}_{S_j},\ \frac{d_j}{(\alpha+\gamma)}\left(\mathbf{X}_{S_j}^T\mathbf{X}_{S_j}\right)^{-1}\right), \quad j = 2, \dots, p, \\
d_j \mid S_j, \mathbf{X}_n &\overset{ind}{\sim} IG\left(\frac{\alpha n + \nu_0}{2},\ \frac{\alpha n}{2}\left(\widehat{d}_{S_j} + \frac{\nu_0'}{n}\right)\right), \quad j = 1, \dots, p, \\
\pi(S_j \mid \mathbf{X}_n) &\propto \pi_j(S_j)\left(1 + \frac{\alpha}{\gamma}\right)^{-\frac{|S_j|}{2}} (\widehat{d}_{S_j})^{-\frac{\alpha n + \nu_0}{2}}, \quad j = 2, \dots, p.
\end{aligned}
$$

An MH algorithm can be used for the posterior inference of S_j.

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Remark and next week

Remark

- The use of ESC prior for posterior inference cannot be performed with the rstan package.
- The main reason is that a discrete prior is used when selecting the support of the Cholesky factor.
- In the near future, R code will be uploaded to GitHub (`https://github.com/leekjstat/ESCprior`).

Next week

Model selection